LAB 7: FUZZING FOR INPUT VALIDATION BUGS

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Project: Car Rental Rate | Python + Hypothesis

Objective

Follow the supplied student handout: run the original application, use Hypothesis to discover edge cases in all three functions, fix the validation defects, and rerun the tests.

Functions tested

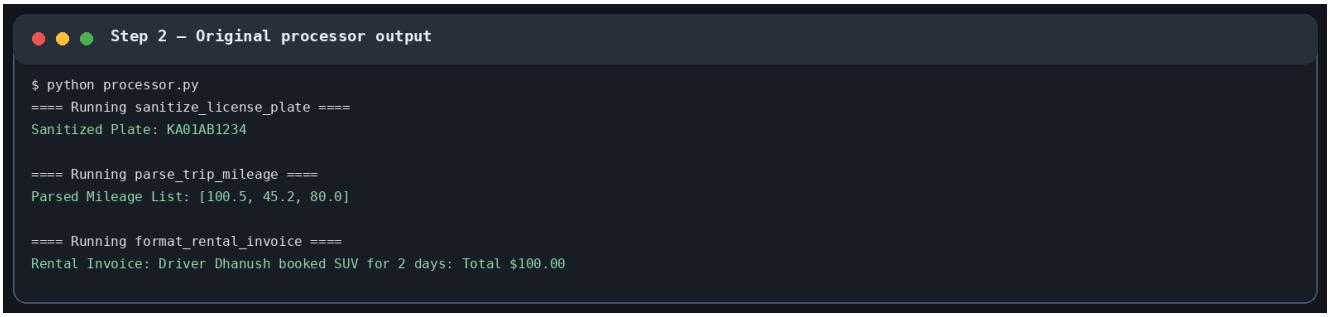
sanitize_license_plate() — normalize a license plate and remove supported delimiters.
parse_trip_mileage() — parse comma-separated mileage values as floats.
format_rental_invoice() — parse Driver:Days:Category and calculate the rental total.

Step 1 — Dependencies

The CI runner installed pytest and Hypothesis from requirements.txt under Python 3.12.16. The test framework and fuzzing library were installed before running the original and fixed versions.

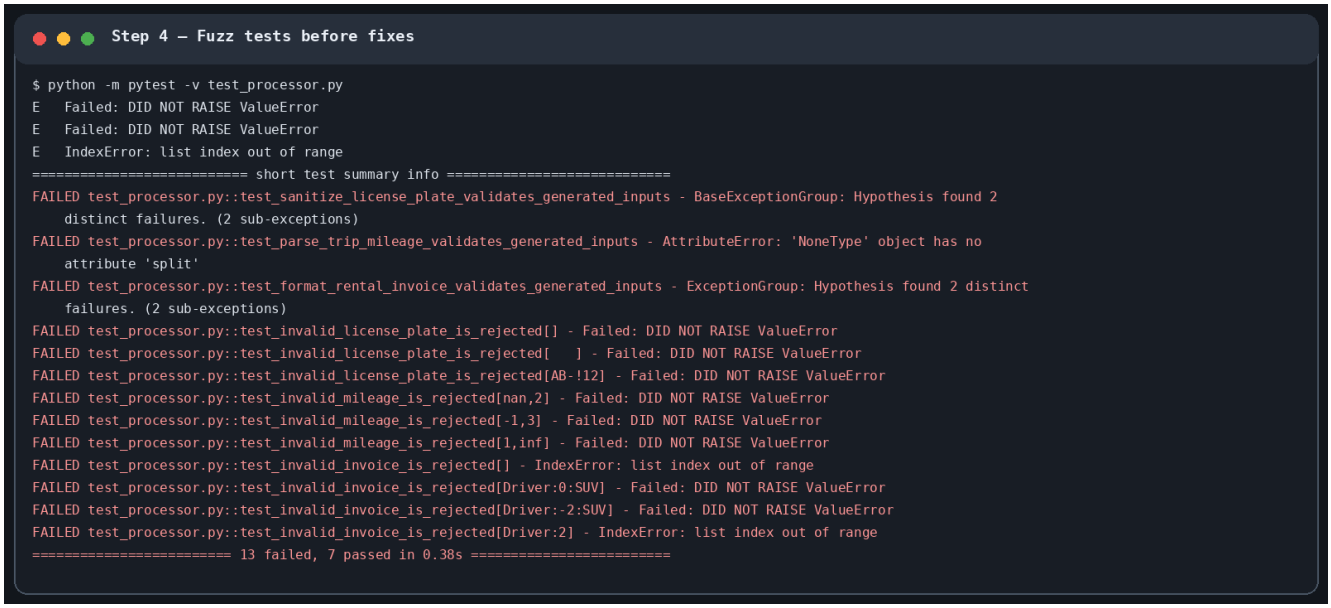
Step 2 — Original application output

The starter program completed with ordinary sample inputs. Its successful normal-path output did not cover malformed or unexpected values.



Steps 3–4 — Hypothesis fuzzing and initial failures

The property-based tests generated strings, None values and integers for each function. Additional regression cases checked empty and invalid plates, empty/nonnumeric/negative/non-finite mileage, malformed invoice records, and non-positive rental days.



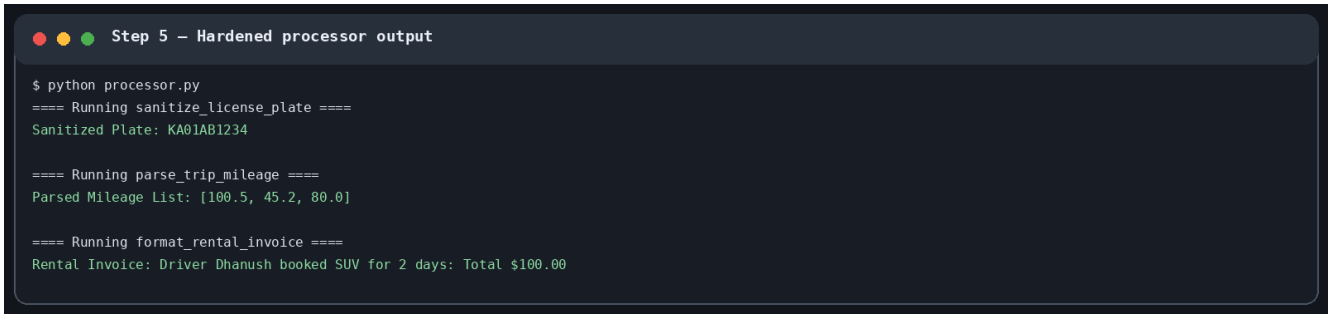
Baseline outcome: 13 failed, 7 passed in 0.38s. The original implementation leaked AttributeError/IndexError and accepted invalid plate characters, negative or non-finite mileage, and zero/negative rental days.

Step 5 — Hardened application

Input types and structure are validated before processing. Wrong types raise TypeError; invalid values raise ValueError rather than leaking incidental parser exceptions.

Fixes applied

• License plates must be non-empty and alphanumeric after supported separators are removed.
• Mileage entries must be numeric, finite and non-negative; empty CSV fields are rejected.
• Rental records must contain exactly three fields, a non-empty driver and category, and a positive integer day count.
• Existing pricing is preserved: SUV $50/day; other categories $30/day.



Step 6 — Final test execution

The same fuzz and regression suite was executed against the fixed processor. The result below is taken from the successful pytest run.

Step 6 — Fuzz tests after fixes

```
$ python -m pytest -v test_processor.py
============================= test session starts ==============================
platform linux -- Python 3.12.16, pytest-9.2.0, pluggy-1.7.0 -- /opt/hostedtoolcache/Python/3.12.16/x64/bin/python
cachedir: .pytest_cache
hypothesis profile 'ci' -> database=None, deadline=None, print_blob=True, derandomize=True,
    suppress_health_check=(HealthCheck.too_slow,)
rootdir: /home/runner/work/SE-Labs-PES1UG25CS812/SE-Labs-PES1UG25CS812/Lab_7_PES1UG25CS812
plugins: hypothesis-6.170.15
collecting ... collected 20 items

test_processor.py::test_sanitize_license_plate_validates_generated_inputs PASSED [  5%]
test_processor.py::test_parse_trip_mileage_validates_generated_inputs PASSED [ 10%]
test_processor.py::test_format_rental_invoice_validates_generated_inputs PASSED [ 15%]
test_processor.py::test_sanitize_license_plate_removes_supported_delimiters PASSED [ 20%]
test_processor.py::test_parse_trip_mileage_returns_float_values PASSED   [ 25%]
test_processor.py::test_format_rental_invoice_uses_suv_rate PASSED       [ 30%]
test_processor.py::test_format_rental_invoice_uses_standard_rate_for_other_categories PASSED [ 35%]
test_processor.py::test_invalid_license_plate_is_rejected[] PASSED       [ 40%]
test_processor.py::test_invalid_license_plate_is_rejected[   ] PASSED    [ 45%]
test_processor.py::test_invalid_license_plate_is_rejected[AB-!12] PASSED [ 50%]
test_processor.py::test_invalid_mileage_is_rejected[] PASSED             [ 55%]
test_processor.py::test_invalid_mileage_is_rejected[1,,2] PASSED         [ 60%]
test_processor.py::test_invalid_mileage_is_rejected[nan,2] PASSED        [ 65%]
test_processor.py::test_invalid_mileage_is_rejected[-1,3] PASSED         [ 70%]
test_processor.py::test_invalid_mileage_is_rejected[1,inf] PASSED        [ 75%]
test_processor.py::test_invalid_invoice_is_rejected[] PASSED             [ 80%]
...

============================== 20 passed in 0.31s ==============================
```

Final outcome: 20 passed in 0.31s. All three Hypothesis fuzz tests and all deterministic regression tests passed.

Conclusion

The before-and-after runs demonstrate that fuzzing exposed real input-validation defects and that the hardened implementation now handles the tested valid and invalid inputs consistently.